

Advanced (Carolina Reaper)

**Q1.** Given  $f(x) = x^2$  and  $g(x) = 2x + 1$ , find  $(f \circ g)(x)$

- (A)  $4x^2 + 4x + 1$
- (B)  $4x^2 + 1$
- (C)  $2x^2 + 1$
- (D)  $x^2 + 2x + 1$
- (E)  $x^2 + 1$

**Q2.** If  $f(x) = \frac{1}{x}$  and  $g(x) = \frac{1}{x+1}$ , what is  $(f \circ g)(x)$ ?

- (A)  $\frac{x+1}{x}$
- (B)  $\frac{x}{x+1}$
- (C)  $\frac{1}{x(x+1)}$
- (D)  $x(x+1)$
- (E)  $\frac{1}{x+1}$

**Q3.** For  $f(x) = |x|$  and  $g(x) = x - 2$ , find  $(f \circ g)(-3)$

- (A) 5
- (B) 3
- (C) 1
- (D) -1
- (E) -3

**Q4.** What is the domain of  $(f \circ g)(x)$  when  $f(x) = \frac{1}{x}$  and  $g(x) = \frac{1}{x-1}$ ?

- (A)  $(-\infty, 1) \cup (1, \infty)$
- (B)  $(-\infty, 1)$
- (C)  $(1, \infty)$
- (D)  $(-\infty, 0) \cup (0, \infty)$
- (E)  $[0, \infty)$

**Q5.** If  $f(x) = x^3$  and  $g(x) = 2x$ , what is  $(f \circ g)(2)$ ?

- (A) 16
- (B) 8
- (C) 4
- (D) 2
- (E) 0

**Q6.** Given  $f(x) = \sqrt{x}$  and  $g(x) = x^2 + 1$ , find  $(f \circ g)(x)$

- (A)  $\sqrt{x^2 + 1}$
- (B)  $\sqrt{x^2}$
- (C)  $x^2 + 1$
- (D)  $\sqrt{x}$
- (E)  $x + 1$

**Q7.** What is the domain of  $(f \circ g)(x)$  when  $f(x) = \sqrt{x}$  and  $g(x) = x - 2$ ?

- (A)  $(-\infty, 2]$
- (B)  $[2, \infty)$
- (C)  $(-\infty, 2)$
- (D)  $(-2, \infty)$
- (E)  $(-2, 2)$

**Q8.** If  $f(x) = 3x$  and  $g(x) = x^2$ , find  $(f \circ g)(-1)$

- (A) 3
- (B) -3
- (C) 0
- (D) 1
- (E) -1

**Open Ended Questions (FRQ)**

**Q9.** Determine the domain and range of  $(f \circ g)(x)$  when  $f(x) = \frac{1}{x^2}$  and  $g(x) = x + 1$ . Provide a clear explanation.

**Q10.** Given  $f(x) = |x - 1|$  and  $g(x) = x - 2$ , evaluate  $(f \circ g)(x)$  for  $x = 0, 1, 2$ . Describe any observations about the behavior of the composite function.

**Chef's Tips**

- Tip 1: Allow students to use calculators for checking their work
- Tip 2: Encourage students to simplify their answers when possible
- Tip 3: Consider providing a formula sheet with key composite function identities